

MAHRANE AMOR

Third-Year Data Science Student

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Education

National School of Electronics and Telecommunications of Sfax (ENET'COM) Sep. 2024 – Present
Data Engineering and Decisional System (Data Science) Sfax, Tunisia

Preparatory Institute for Engineering Studies of Kairouan (IPEIK) Sep. 2022 – May 2024
Preparatory Cycle Technology and Physics Track Kairouan, Tunisia

Professional Experience

YOUR OS Jul 2026 – August 2026

AI Engineer Intern–Intelligent Medical Assistant Remote

- Designed an Agentic RAG architecture with LangGraph, enabling context-aware medical reasoning across patient records and external medical knowledge.
- Engineered a medical retrieval pipeline combining hybrid search, reranking, corrective RAG (CRAG), query routing, and specialty-aware retrieval for grounded responses.
- Conducted comparative evaluations of open-source medical LLMs and Speech-to-Text models, including MedGemma, Llama, Qwen, Whisper Large-v3, and faster-whisper, across performance and integration criteria.
- Integrated faster-whisper into the agentic workflow, transforming spoken clinical queries into structured textual inputs for downstream query classification, retrieval, and LLM-based reasoning.
- Tools:** Python, FastAPI, LangGraph, OpenAI, Mistral AI, Hugging Face, PostgreSQL, Qdrant, MinIO, Redis, Docker, GitHub Actions, Next.js, TypeScript.

BHIVES.AI July 2025 – August 2025

AI Engineer Intern–Fine-Tuning F5-TTS Model for Arabic Neural Speech Generation Remote

- Implemented and fine-tuned the F5-TTS model for Arabic speech generation.
- Applied progressive training strategies to improve model stability and audio quality.
- Optimized inference performance for smoother and natural speech output.
- Evaluated generated speech using technical metrics and perceptual listening tests.
- Tools:** PyTorch, F5-TTS, Hugging Face, TorchAudio, NumPy, Google Colab, Hugging Face

KEYRUS June 2025 – June 2025

Data Analyst Intern–Retail Business Intelligence and Performance Analytics Remote

- Designed and implemented decision-support dashboards for commercial performance monitoring in the retail sector.
- Built interactive dashboards for inventory management, sales revenue analysis, and customer satisfaction tracking.
- Optimized inference performance for smoother and natural speech output.
- Modeled and integrated data from multiple sources to ensure accurate and consistent KPIs.
- Developed advanced metrics and calculated measures using DAX.
- Deployed dashboards and data solutions on Microsoft Azure for secure access and scalability.
- Tools:** SQL Server (SSMS), Power BI, Microsoft Azure.

Projects

Real-Time Intelligent Monitoring and Anomaly Detection in 5G Networks January 2026-April 2026

- Designed and implemented a real-time data pipeline for monitoring 5G network performance using streaming technologies and performed exploratory data analysis (EDA) to identify key factors affecting network anomalies.
- Integrated data processing components using Apache Kafka for real-time data ingestion and communication.
- Developed machine learning and Deep Learning models to detect anomalies such as network failures and security threats exposed results through REST APIs and interactive dashboards..
- Visualized network insights and anomalies using Power BI for decision support and monitoring.
- Tools:** Python, Apache Kafka, FastAPI, PostgreSQL, Power BI, Scikit-learn / PyTorch, Docker.

- Developed 2D and 3D U-Net deep learning models for multi-class brain tumor segmentation from multimodal MRI scans using PyTorch.
- Engineered the medical imaging preprocessing pipeline, including NIfTI processing, z-score normalization, spatial cropping, depth resizing, and volumetric data preparation.
- Optimized model training with Dice + Cross-Entropy loss, cosine learning-rate scheduling, and mixed-precision training (AMP), with patient-level data splitting to prevent data leakage.
- Evaluated segmentation performance using Dice scores across Whole Tumor, Tumor Core, and Enhancing Tumor regions, and deployed inference through a FastAPI interface.
- **Tools:** Python, PyTorch, 3D U-Net, NumPy, NIfTI, CUDA/AMP, FastAPI, Uvicorn.

Technical Skills

Programming Languages:	Python, R, Java, C, JavaScript.
Big Data	:Scala, SQL (MySQL, PostgreSQL, Oracle), NoSQL (MongoDB).
AI and Machine Learning:	Agentic RAG, LLM, Fine-tuning, Computer Vision, MLOps, TensorFlow, PyTorch.
Operating Systems	:Windows, Linux.
DevOps and Deployment	:Git, GitHub, Docker, FastAPI, Azure.
Development Tools	: VS Code, Antigravity, Eclipse.

Certifications

Microsoft:	Azure Data Fundamentals(DP-900).
Udemy:	Machine learning A-Z.
Cognitive Class :	Python for Data Science
HackRank:	Problem solving.
Sololearn:	SQL intermediate
DataCamp:	AI Fundamentals.
365 Data Science:	Probability.
365 Data Science:	Git And Github.

Community Involvement

Participant at MenaML winter school of Machine Learning 2026	January 2026
Participant	KAUST University
• Expanded professional network with AI researchers and industry experts, and enhanced understanding of cutting-edge ML research trends.	
Actif Member at Google Developer Student Club (GDSC)	September 2024-Present
Actif Member	ENETCOM
• Participated in the MindShift Hackathon, contributing to AI/ML project development and collaborative problem-solving.	
Actif Member at the Junior Chamber International (JCI)	September 2021-August 2024
Actif Member	JCI Bennene
• Participated in community projects and leadership initiatives, developing project management, teamwork, and communication skills while contributing to social impact activities.	

Languages

* English [Fluent]	* French [Fluent]	* Arabic [Native]	* German [A2]
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